Algebra and Discrete Mathematics (ADM)

Tutorial 8 Eulerian tours

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Graph

Definition

A *graph* consists of two sets: $V(G)$, called the *vertex set*, and $E(G)$, called the edge set. An *edge*, denoted xy, is an unordered pair of vertices $x, y \in V(G)$, where x and y may be the same.

Theorem (Handshaking Lemma)

Let $G = (V, E)$ be a finite graph. Suppose $V = \{v_1, v_2, \ldots, v_n\}$, then

$$\deg(v_1) + \deg(v_2) + \cdots + \deg(v_n) = 2|E|$$

Draw a graph

Question

Draw a graph with 6 vertices such that their degrees are

$$1,\ 2,\ 3,\ 4,\ 5,\ 2$$

We have

$$\sum_{v} \deg(v) = 1 + 2 + 3 + 4 + 5 + 2 = 17,$$

which is odd. Hence, such a graph does not exist.

Draw a graph

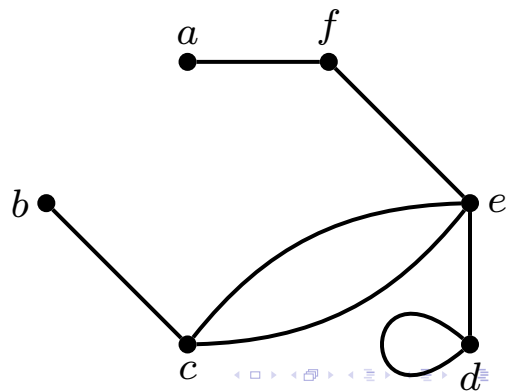
Question

Draw a graph with 6 vertices such that their degrees are

$$1,\ 1,\ 3,\ 3,\ 4,\ 2$$

$\deg(a) = 1, \quad \deg(b) = 1, \quad \deg(c) = 3,$

$\deg(d) = 3, \quad \deg(e) = 4, \quad \deg(f) = 2,$



Draw a graph

Question

Construct a graph to represent a club with 13 members, where each member is acquainted with exactly 5 other members.

The sum of degrees is given by

$$13 \times 5 = 65$$

which is odd. Hence, such a graph does not exist.

Non-existence of certain regular graphs

Statement

Prove that a k-regular graph, where k is odd, with an odd number of vertices does not exist.

Proof.

If a graph has n vertices and is k-regular, then the sum of all vertex degrees is

$$nk.$$

Since both n and k are odd, the product nk is odd.
But by the Handshaking Lemma, the sum of the degrees of all vertices must be even.
This is a contradiction, so such a graph cannot exist. □

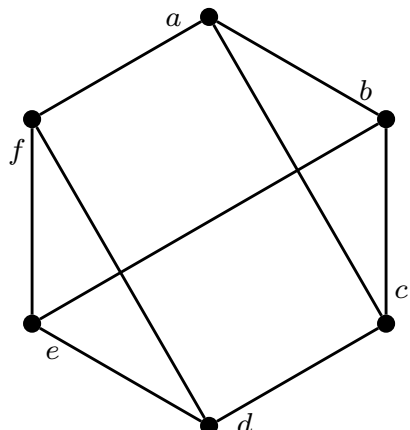
Draw a graph

Question

Draw a graph with 6 vertices, each of degree 3

$$\frac{6 \times 3}{2} = 9$$

so the graph must have 9 edges.



Draw a graph

Question

Draw a graph with 7 vertices, each of degree 3

$$7 \times 3 = 21,$$

which is odd. Hence, such a graph does not exist.

Draw a graph

Question

Draw a graph with 4 vertices, each of degree 1



Note that this graph is not connected.

Draw a graph

Question

Draw a graph with 6 edges, 4 vertices with degrees

$$1,\ 2,\ 3,\ 4$$

It must hold that

$$2|E| = \sum_{v \in V(G)} \deg(v)$$

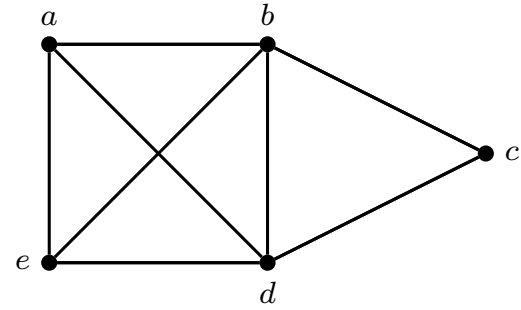
However

$$2|E| = 2 \times 6 = 12$$

$$\sum_{v \in V(G)} \deg(v) = 1 + 2 + 3 + 4 = 10$$

Hence, such a graph does not exist.

Touring a graph



- Path: does not repeat vertices
  - Example: $a,\ b,\ c,\ d,\ e$
- Trail: does not repeat edges
  - Example: $a,\ b,\ e,\ a,\ d,\ c$
- Cycle: a closed path
  - Example: $a,\ b,\ c,\ d,\ e,\ a$
- Circuit: a closed trail
  - Example: $a,\ b,\ c,\ d,\ b,\ e,\ a$

Eulerian and semi-Eulerian

Theorem

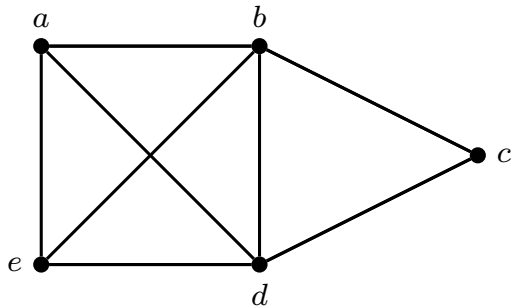
A graph G is Eulerian if and only if

- *G is connected and*
- *every vertex has an even degree*

A graph G is semi-Eulerian if and only if

- *G is connected and*
- *exactly two vertices have odd degree*

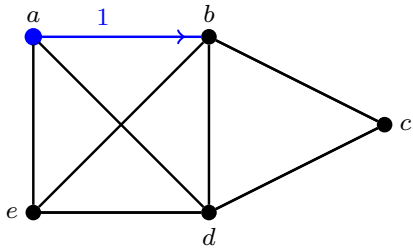
Semi-Eulerian graph

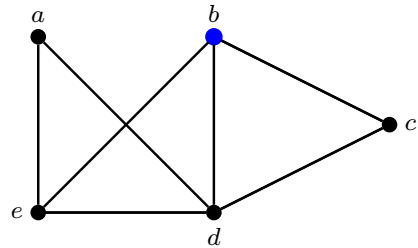


$$\deg(a) = 3, \quad \deg(b) = 4, \quad \deg(c) = 2, \quad \deg(d) = 4, \quad \deg(e) = 3$$

The graph is semi-Eulerian

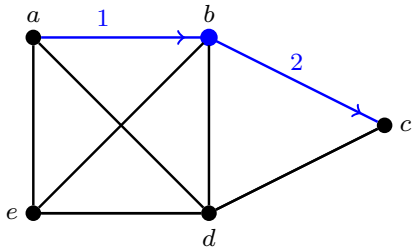
Fleury's Algorithm

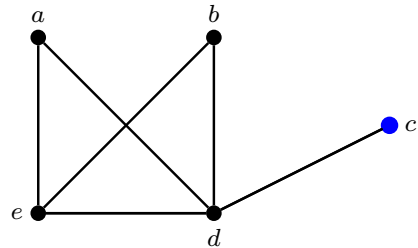




- Step 1. Start at vertex a
- Step 2. Choose edge ab
- Step 3. Travel to vertex b and delete edge ab. Current vertex: b.

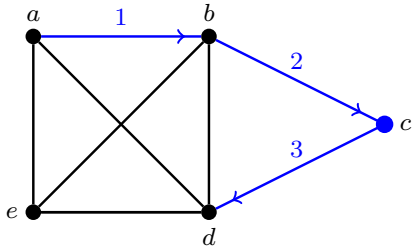
Fleury's Algorithm

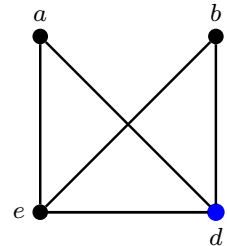




- Step 2. Choose edge bc
- Step 3. Travel to vertex c and delete edge bc. Current vertex: c.

Fleury's Algorithm

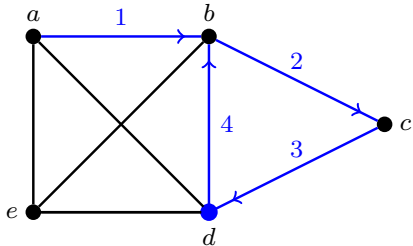


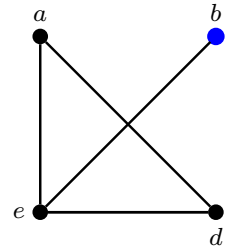




- Step 2. The only available choice is edge cd.
- Step 3. Travel to vertex d and delete edge cd. Current vertex: d.

Fleury's Algorithm

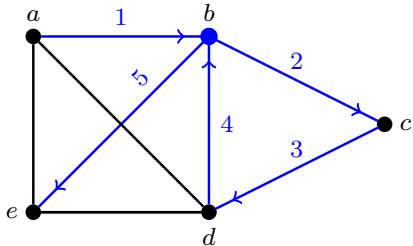


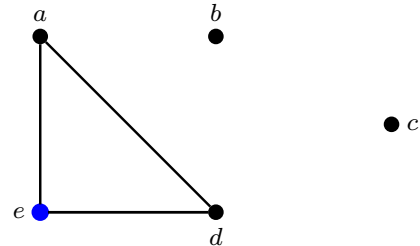




- Step 2. Choose edge db
- Step 3. Travel to vertex b and delete edge db. Current vertex: b.

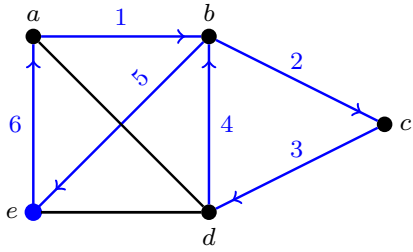
Fleury's Algorithm

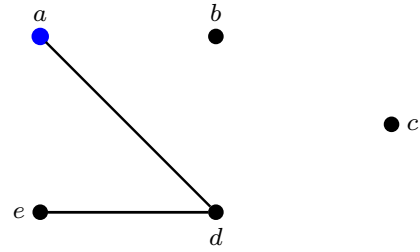




- Step 2. The only available choice is be
- Step 3. Travel to vertex e and delete edge be. Current vertex: e.

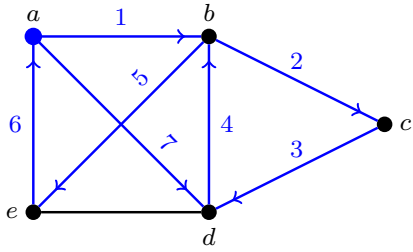
Fleury's Algorithm

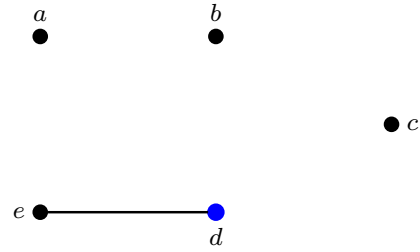




- Step 2. Choose ea
- Step 3. Travel to vertex a and delete edge ea. Current vertex: a.

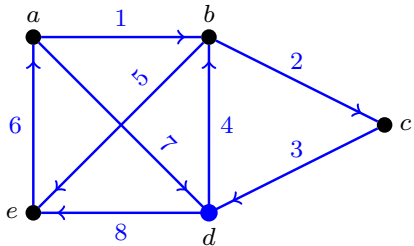
Fleury's Algorithm

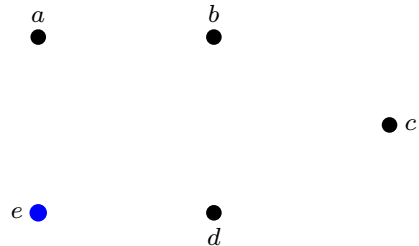




- Step 2. The only available choice is ad
- Step 3. Travel to vertex d and delete edge ad. Current vertex: d.

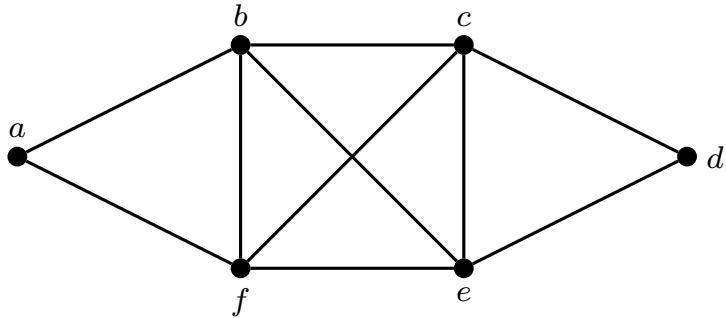
Fleury's Algorithm





- Step 2. The only available choice is de
- Step 3. Travel to vertex e and delete edge de.
- We have obtained an Eulerian trail

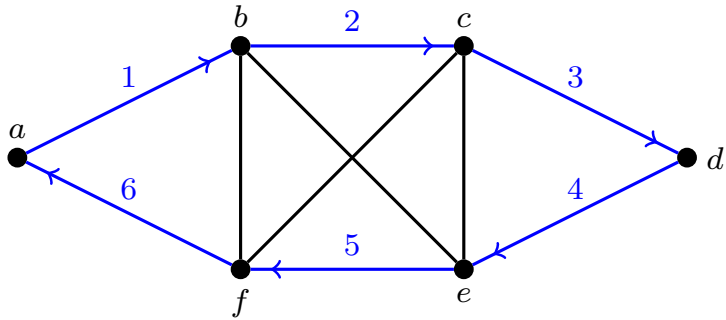
Hierholzer's Algorithm



$\deg(a) = 2, \quad \deg(b) = 4, \quad \deg(c) = 4, \quad \deg(d) = 2, \quad \deg(e) = 4, \quad \deg(f) = 4$

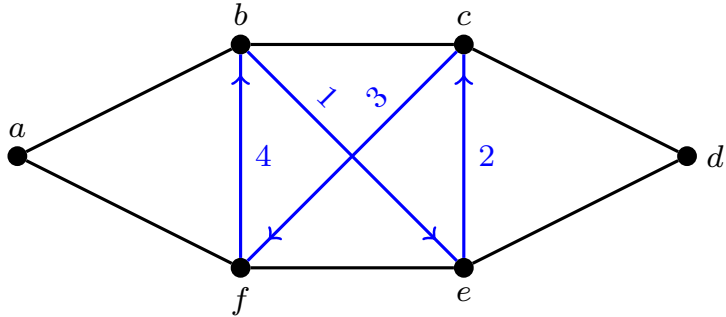
The graph is Eulerian.

Hierholzer's Algorithm



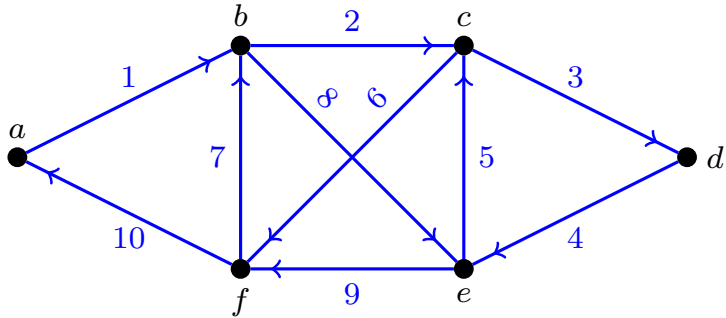
- Step 1. Choose vertex a and find a circuit starting at a.

Hierholzer's Algorithm



- Step 2. Since $\deg(b) = 4$, two unused edges remain incident to b, so we can find a second circuit starting at b.

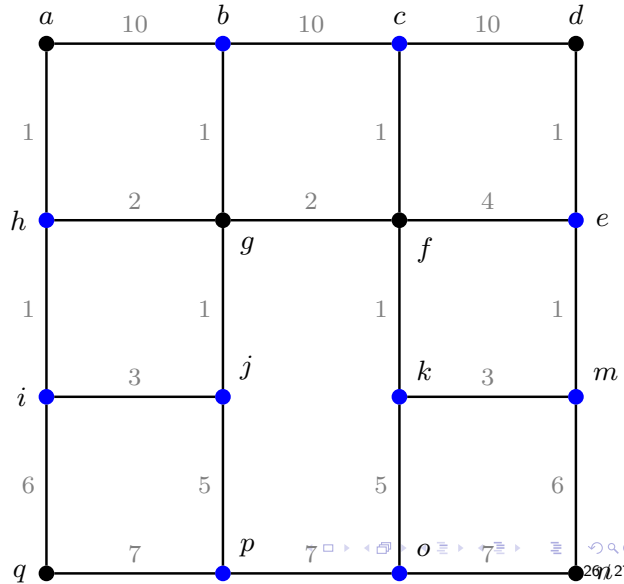
Hierholzer's Algorithm



- Step 3. Combine the new circuit with the existing one.
- We have obtained an Eulerian circuit.

Chinese Postman Problem

- Find an optimal Eulerization, where the edge weights represent cost.
- The unweighted version suggests first pairing odd vertices using as few duplicated edges as possible.
- We first attempt to pair vertices along paths using edges of weight 1 as much as possible
- We are able to do this for all vertices except o and p, and we duplicate edge op



Chinese Postman Problem

- The graph has 10 odd vertices, so at least 5 edge duplications are needed.
- If we avoid duplicating op, then o and p each require a duplicated incident edge of weight at least 5. The other duplications contribute at least 3, so the total added weight is at least 13.
- If we duplicate op, its weight is 7. Among the remaining 8 odd vertices, only hi and em are adjacent via edges of weight 1. Thus, at least two pairs require paths of weight at least 2, giving an additional weight of at least $1 + 1 + 2 + 2 = 6$.
- Therefore, the Eulerization shown, with total added weight 13, is optimal.

